

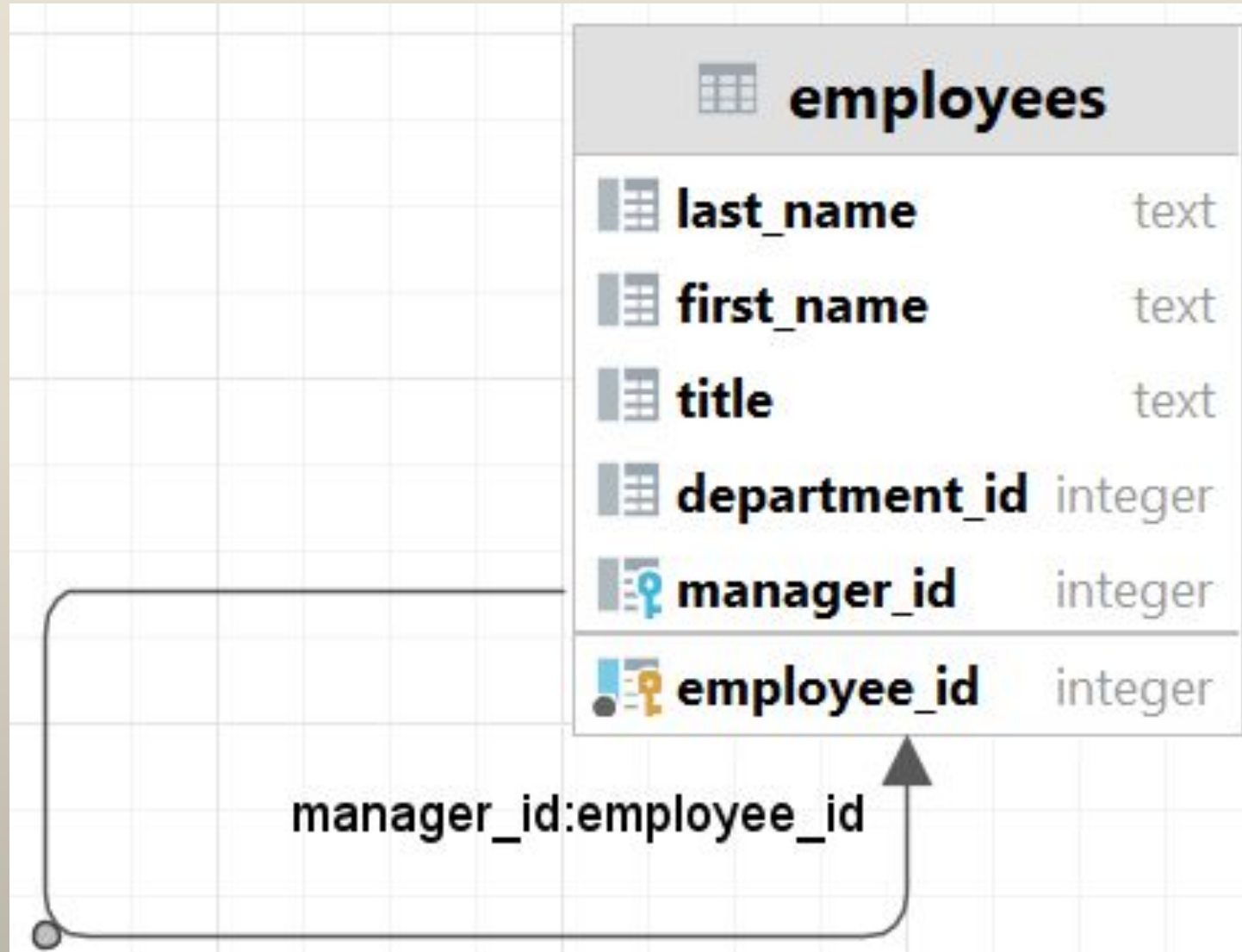


Self Joins

A **self join** is a regular join, but the table is joined with itself.

```
create schema if not exists self_join;
create table if not exists self_join.employees (
    employee_id int
    , last_name text
    , first_name text
    , title text
    , department_id int
    , manager_id int -- the employee_id for the manager
    constraint manager_fk
    references self_join.employees(employee_id)
    deferrable initially deferred
    , constraint employees_pk primary key (employee_id)
);
```

Self Joins

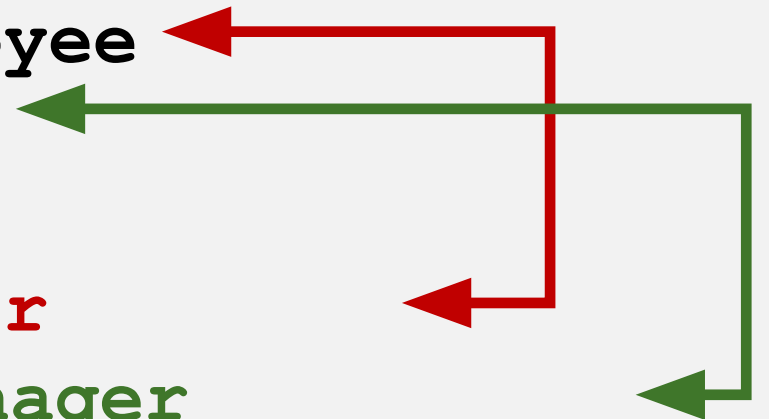


```
insert into self_join.employees values
  (1, 'Jones', 'Mark', 'Department Chair', 2, null)
, (2, 'Jones', 'Elmer', 'Instructor', 4, 1)
, (3, 'Li', 'Jingke ', 'Professor', 2, 2)
, (4, 'Hernandez', 'Olivia', 'Instructor', 1, 9)
, (5, 'Feng', 'Wu-chang', 'Professor', 2, 1)
, (6, 'Watson', 'Denise', 'Instructor', 6, 8)
, (7, 'Hardy', 'Thomas', 'Instructor', 5, 2)
, (8, 'O' 'Leary', 'Rhea', 'Professor', 4, 9)
, (9, 'Locario', 'Paulo', 'Professor', 6, 1)
, (10, 'Chaney', 'Jesse', 'Senior Instructor', null, 11)
, (11, 'McGrath', 'Kevin', 'Senior Instructor', null, 10)
, (12, 'Wilson', 'Mike', 'Senior Instructor', 1, null)
;
```

```
select count(*)
from self_join.employees;
12 rows
```

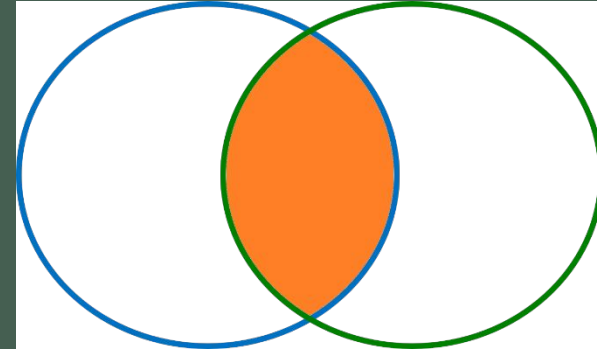
Self Joins

```
SELECT worker.first_name as employee
      , manager.first_name as boss
      , worker.title
FROM self_join.employees as worker
     JOIN self_join.employees as manager
       ON worker.manager_id = manager.employee_id
ORDER BY boss, employee;
```

A diagram illustrating the self-join operation. Two arrows originate from the 'employees' table in the FROM clause. A red arrow points to the 'employee' alias in the SELECT clause, and a green arrow points to the 'boss' alias. A second set of red and green arrows originates from the 'employees' table in the JOIN clause and points to the 'employee' and 'boss' aliases respectively, indicating the mapping of the joined table to the aliases.

Let's do an inner join on the table with itself, a self-join.

```
SELECT worker.first_name as employee
      , manager.first_name as boss
      , worker.title
FROM self_join.employees as worker
  JOIN self_join.employees as manager
    ON worker.manager_id = manager.employee_id
ORDER BY boss desc, employee;
```

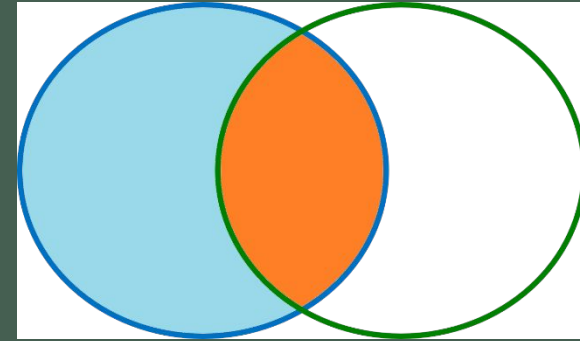


employee	boss	title
Denise	Rhea	Instructor
Olvia	Paulo	Instructor
Rhea	Paulo	Professor
Elmer	Mark	Instructor
Paulo	Mark	Professor
Wu-chang	Mark	Professor
Jesse	Kevin	Senior Instructor
Kevin	Jesse	Senior Instructor
Jingke	Elmer	Professor
Thomas	Elmer	Instructor

10 rows

```

SELECT worker.first_name as employee
       , manager.first_name as boss
       , worker.title
FROM self_join.employees as worker
     left JOIN self_join.employees as manager
     ON worker.manager_id = manager.employee_id
ORDER BY boss desc, employee;
  
```



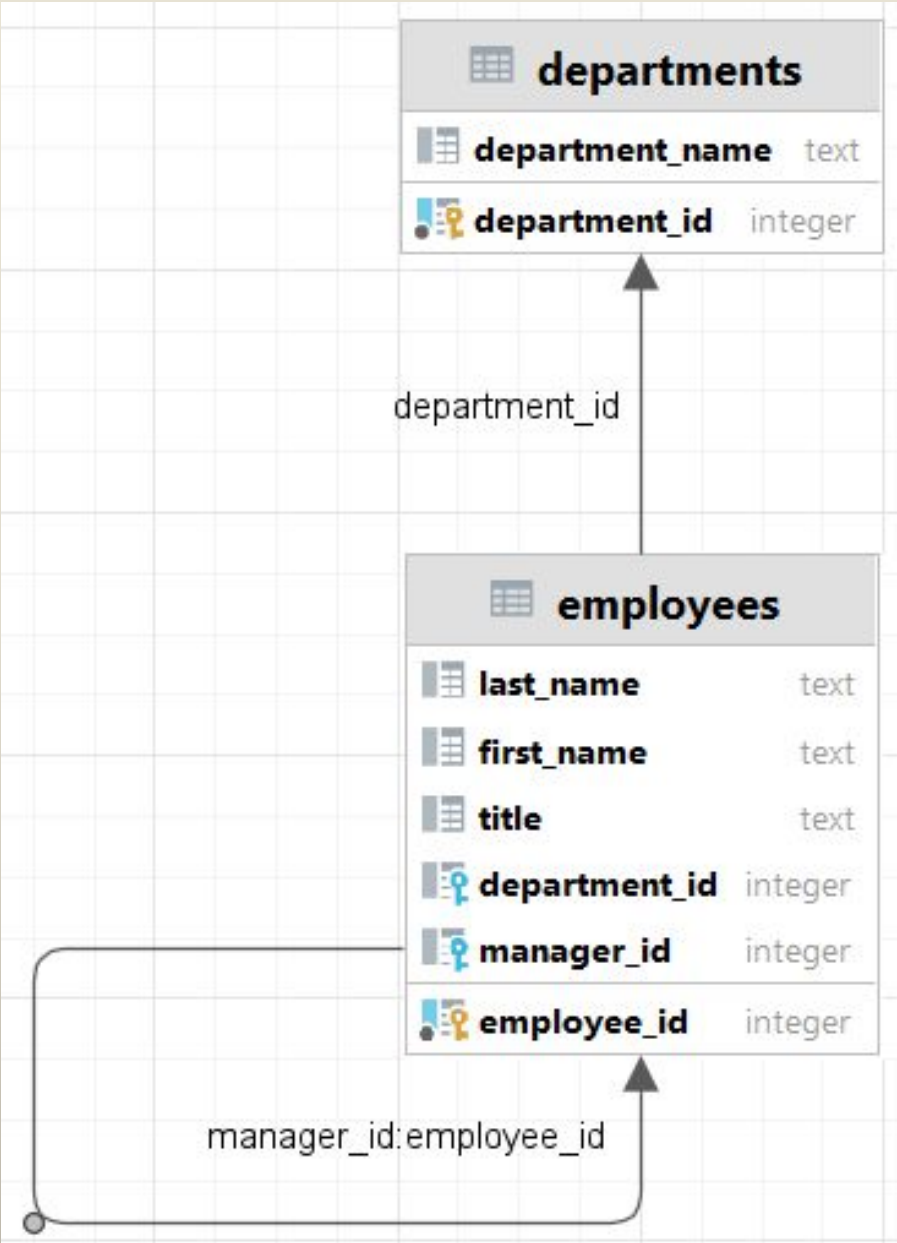
employee	boss	title
Mark		Department Chair
Mike		Senior Instructor
Denise	Rhea	Instructor
Olvia	Paulo	Instructor
Rhea	Paulo	Professor
Elmer	Mark	Instructor
Paulo	Mark	Professor
Wu-chang	Mark	Professor
Jesse	Kevin	Senior Instructor
Kevin	Jesse	Senior Instructor
Jingke	Elmer	Professor
Thomas	Elmer	Instructor

12 rows

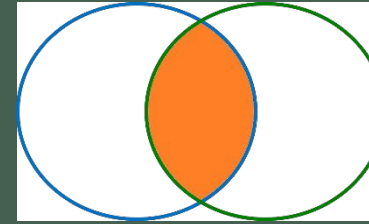
```
create table if not exists self_join.departments (  
    department_id int  
    , department_name text  
    , constraint departments_pk  
      primary key (department_id)  
);
```

```
insert into self_join.departments values  
    (2, 'Executive ')  
    , (1, 'IT')  
    , (4, 'Accounting')  
    , (6, 'Customer Service')  
    , (5, 'Sales')  
    , (3, 'Manufacturing')  
;
```

```
create table if not exists self_join.employees (  
    employee_id int  
    , last_name text  
    , first_name text  
    , title text  
    , department_id int  
    , manager_id int    -- the employee_id for the manager  
    constraint manager_fk  
        references self_join.employees(employee_id)  
        deferrable initially deferred  
    , constraint employees_pk primary key (employee_id)  
    , constraint dept_fk foreign key (department_id)  
        references self_join.departments(department_id)  
);
```

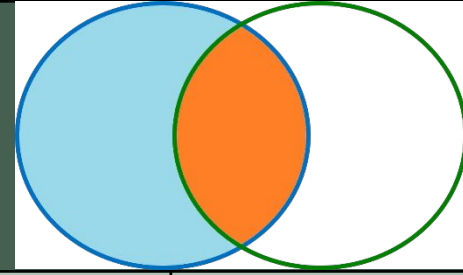


```
select e.first_name, e.last_name, e.title, department_name
from self_join.employees as e
  join self_join.departments as d
    on e.department_id = d.department_id;
```



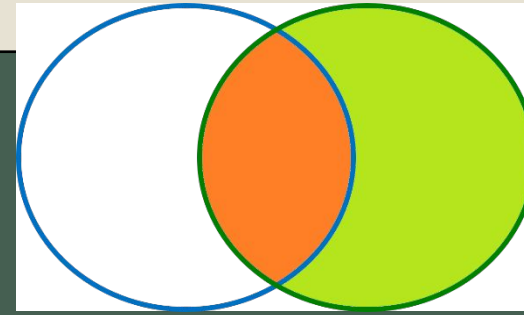
first_name	last_name	title	department_name
Mark	Jones	Department Chair	Executive
Elmer	Jones	Instructor	Accounting
Jingke	Li	Professor	Executive
Olvia	Hernandez	Instructor	IT
Wu-chang	Feng	Professor	Executive
Denise	Watson	Instructor	Customer Service
Thomas	Hardy	Instructor	Sales
Rhea	O'Leary	Professor	Accounting
Paulo	Locario	Professor	Customer Service
Mike	Wilson	Senior Instructor	IT

```
select e.first_name, e.last_name, e.title, department_name
from self_join.employees as e
  left join self_join.departments as d
    on e.department_id = d.department_id;
```



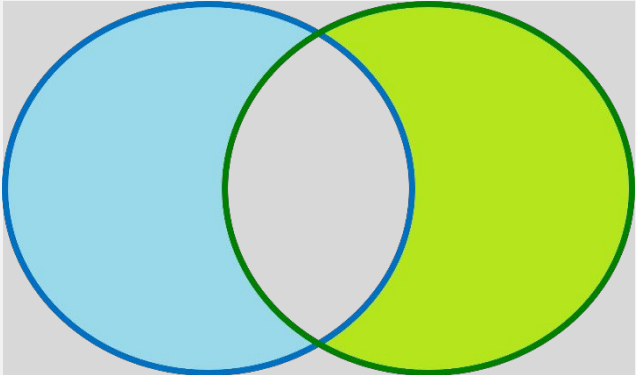
first_name	last_name	title	department_name
Mark	Jones	Department Chair	Executive
Elmer	Jones	Instructor	Accounting
Jingke	Li	Professor	Executive
Olvia	Hernandez	Instructor	IT
Wu-chang	Feng	Professor	Executive
Denise	Watson	Instructor	Customer Service
Thomas	Hardy	Instructor	Sales
Rhea	O'Leary	Professor	Accounting
Paulo	Locario	Professor	Customer Service
Jesse	Chaney	Senior Instructor	
Kevin	McGrath	Senior Instructor	
Mike	Wilson	Senior Instructor	IT

```
select e.first_name, e.last_name, e.title, department_name
from self_join.employees as e
  right join self_join.departments as d
    on e.department_id = d.department_id;
```



first_name	last_name	title	department_name
Mark	Jones	Department Chair	Executive
Elmer	Jones	Instructor	Accounting
Jingke	Li	Professor	Executive
Olvia	Hernandez	Instructor	IT
Wu-chang	Feng	Professor	Executive
Denise	Watson	Instructor	Customer Service
Thomas	Hardy	Instructor	Sales
Rhea	O'Leary	Professor	Accounting
Paulo	Locario	Professor	Customer Service
Mike	Wilson	Senior Instructor	IT
			Manufacturing

```
select e.first_name, e.last_name, department_name
from self_join.employees as e
  full join self_join.departments as d
  on e.department_id = d.department_id
where department_name is null
  or last_name is null;
```



first_name	last_name	department_name
Jesse	Chaney	
Kevin	McGrath	
		Manufacturing

